Zero-Shot Hyperspectral Pansharpening Using Hysteresis-Based Tuning for Spectral Quality Control

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***Abstract*— Hyperspectral (HS) pansharpening has received much attention in recent years due to technological and methodological advances that open the door to new application scenarios. However, research on this topic is only now gaining momentum. The most popular methods are still borrowed from the more mature field of multispectral pansharpening and often overlook the unique challenges posed by HS data fusion, such as: 1) the very large number of bands; 2) the overwhelming noise in selected spectral ranges; 3) the significant spectral mismatch between panchromatic (PAN) and HS components; and 4) a typically high resolution ratio. Imprecise data modeling especially affects spectral fidelity. Even state-of-the-art (SotA) methods perform well in certain spectral ranges and much worse in others, failing to ensure consistent quality across all bands, with the risk of generating unreliable results. Here, we propose an HS pansharpening method that explicitly addresses this problem and ensures uniform spectral quality. To this end, a single lightweight neural network is used, with weights that adapt on the fly to each band. During fine-tuning, the spatial loss is turned on and off to ensure a fast convergence of the spectral loss to the desired level, according to a hysteresis-like dynamic. Furthermore, the spatial loss itself is appropriately redefined to account for nonlinear dependencies between PAN and spectral bands. Overall, the proposed method is fully unsupervised, with no prior training on external data, flexible, and low-complexity. Experiments on a recently published benchmarking toolbox show that it ensures excellent sharpening quality, competitive with the SotA, consistently across all bands.**

Received 24 January 2025; revised 23 April 2025 and 10 June 2025; accepted 24 June 2025. Date of publication 27 June 2025; date of current version 3 July 2025. This work was supported in part by the Italian Space Agency (ASI) "Space It Up!" Spoke3, Codice Unico di Progetto (CUP), under Grant E63C24000220006; and in part by the National Recovery and Resilience Plan (NRRP), Mission 4 Component 2 Investment 1.4—call for tender no. 3138 of 16 December 2021, rectified by Decree n. 3175 of 18 December 2021 of the Italian Ministry of University and Research funded by the European Union—NextGenerationEU; Project code CN_00000033, Concession Decree no. 1034 of 17 June 2022 adopted by the Italian Ministry of University and Research, CUP, Project title "National Biodiversity Future Center—NBFC," under Grant B83C22002930006. *(Corresponding author: Matteo Ciotola.)*

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Digital Object Identifier 10.1109/TGRS.2025.3583877

The software code and the full set of results are shared online on `https://github.com/giu-guarino/rho-PNN`

***Index Terms*— Hyperspectral (HS) images, image fusion, pansharpening, self-supervised learning, spectral unmixing, zero-shot learning.**

I. Introduction

HYPERSPECTRAL (HS) remote sensing imagery is a key resource for monitoring and managing such diverse environments as forests [1], [2], agricultural areas [3], [4], oceans [5], [6], coastal areas [7], [8], inland waters and wetlands [9], [10], snow and ice [11], [12], atmosphere [13], [14], mineral resources [15], [16], [17], and urban areas [18], [19]. Due to the technological constraints of the sensors and the very high number of spectral bands acquired, HS images have a rather low spatial resolution. A cost-effective solution to overcome this limitation is to equip the flight system with an additional camera to acquire, in synchrony with the HS image, a single high-resolution panchromatic (PAN) band. Then, using a fusion process known as pansharpening, the PAN-HS pair is merged into a single high-resolution HS data cube. This data fusion process improves spatial resolution while preserving spectral coherence, thereby boosting the performance of downstream tasks such as classification [20] and land-use/cover mapping [21]. Examples of satellite-borne missions that feature the joint acquisition of PAN and HS images are *PRecursore IperSpettrale della Missione Applicativa* (PRISMA) by the Italian Space Agency (ASI) and Earth Observing-1/Advanced Land Imaging (EO-1/ALI) by the U.S. National Aeronautics and Space Administration (NASA). In both cases, the spatial resolution is 30 m for the HS component and much better for the PAN image, 5 m for PRISMA, and 10 m for EO-1/ALI. Needless to say, the PRISMA case is certainly more interesting, as it promises to provide high-resolution data, but also more challenging since it requires the pansharpening algorithm to handle a resolution ratio of 6 between the available HS data in input and the desired output.

Pansharpening is not a new topic. It has been the object of intense research in the last decades with reference to multispectral (MS) images, where only a limited number of spectral

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bands (from 4 to 8) are available in the visible to near-infrared spectrum. A large number of methods have been proposed for MS pansharpening [22], [23], with very different approaches such as component substitution (CS) [24], [25], [26], [27], multiresolution analysis (MRA) [28], [29], [30], variational optimization [31], [32], [33], [34], and machine/deep learning (DL) [35], [36], [37], [38], [39].

Only recently has the attention shifted toward the HS case, as testified by the growing number of research papers (see [40] for a review) and by open challenges such as [41].

Clearly, MS and HS pansharpening share a common formulation, which may suggest that PAN–MS fusion methods can be readily extended to operate also in the PAN–HS case. Indeed, early methods for HS pansharpening [42] were simple adaptations from the MS case, based on Bayesian models [17], [43], [44], [45], matrix factorization [46], [47], [48], variational optimization [49], component substitution, and MRA [50] (see [51] for a detailed review). However, this approach is overly simplistic, since the HS case has numerous peculiar features that make it particularly challenging.

1) PAN and MS images span comparable spectral ranges, and the PAN bandwidth covers most MS bands. The HS image, instead, has a much larger spectral extension, and most of the HS bands are not covered by the PAN. Therefore, their spatial structure cannot be reliably predicted from it.
2) HS bands are much narrower than MS ones, in the order of 10 nm. This is why, to harvest more energy, larger ground cells are necessary. Even so, the acquired data are rather noisy, with an extremely variable quality, and some groups of bands, in spectral ranges where the sunlight is weaker, must be discarded altogether.
3) Sometimes, the PAN–MS resolution ratio is rather large (e.g., six for PRISMA), hence the pansharpening algorithm is called upon to recover a large amount of missing information.
4) HS images are huge, hence the computational complexity is a serious issue, whatever the optimization strategy adopted. This is especially troublesome for supervised machine learning methods that require large training sets covering the whole data space.

In recent years, a few model-based methods specifically designed for HS pansharpening were proposed, relying on guided filtering [52], variational optimization [53], [54], and saliency-based component substitution [55]. However, the DL revolution has by now spread to the field of remote sensing [56] and, since the seminal 2016 work of Masi et al. [35], DL is a *de facto* standard for pansharpening. Focusing on the HS case, in [57] and [58], dedicated convolutional neural networks (CNNs) were designed to strengthen the spectral prediction capability. In [59], a dual-attention residual network was proposed, with a deep HS image prior module for HS upsampling. In [60] and [61], the problem of scale and resolution variability was tackled using a CNN with arbitrary-scale attention modules [61]. An overcomplete residual network that learns high-level features using constrained receptive fields was proposed in [62]. Multibranch network architectures were also explored in several recent works [63], [64], [65]. Other methods adopted classical pansharpening paradigms and used DL modules to estimate the model parameters [66] or to process the extracted features [67]. All these methods rely on supervised learning and obviate the absence of a real ground truth (GT) by training on reduced-resolution simulated images. The performance is indeed very good on reduced-resolution test data, but drops significantly when moving to the real target of the process, the full-resolution images. In other words, models trained in the subsampled domain do not generalize well to the full-resolution domain. Recently, to overcome these problems, a paradigm shift toward full-resolution training is taking place for both MS [68], [69], [70] and HS [71], [72], [73] pansharpening, with novel learning strategies and suitably defined loss functions. Focusing on the HS case, the RE-RANet method proposed in [71] leverages the ratio image as a latent variable and derives spatial and spectral information of the fused image from the full-resolution PAN and HS components. The R-PNN model [72] performs pansharpening sequentially for each individual band using an unsupervised target-adaptive solution. A divide-and-conquer strategy is used by PCA-Z-PNN, where the HS stack is split into two coherent substacks, which are transformed by principal component analysis and pansharpened individually by the MS-oriented Z-PNN [69] model. A common trait of these three methods is the use of two-term losses that suitably balance spectral and spatial quality.

Aiming to provide an objective overview of recent progress in the field and to support further research, we have recently developed a benchmarking toolbox[1] for HS pansharpening [40]. A large number of state-of-the-art (SotA) methods, both model-based and data-driven, have been reimplemented using a common development environment and evaluated under uniform conditions on a large and representative set of publicly available images. In [40], we performed a comparative analysis of the state of the art. However, the toolbox can be used to perform many more experiments and investigate specific aspects of pansharpening, such as the strengths and weaknesses of different approaches. In particular, here we focus on the issue of nonuniform quality along the spectral dimension, a problem largely neglected in the literature. Indeed, we observed that nearly all methods perform well in some spectral intervals and much worse in others, providing unreliable results in some spectral bands. Of course, spectral richness is a key distinctive feature of HS images, but it becomes meaningless in the absence of adequate guarantees on the accuracy of the observed data. Only under these conditions, these images can be smoothly used by the end user. It should be noted that a major cause of this undesirable behavior is the lack of a quality assessment protocol specifically tailored to the HS case. The widely used performance metrics are simple adaptations of those designed for the very different MS case and, as such, they fail to capture subtle but important differences.

Motivated by the above considerations, in this work, we propose a novel DL-based HS pansharpening method which can keep the spectral distortion under control in each single band,

[1] https://github.com/matciotola/hyperspectral_pansharpening_toolbox

so as to ensure uniform (and top) quality along the whole spectral range. To this end, we build upon the recently proposed [72] rolling pansharpening neural network (rolling PNN or R-PNN) method, whose structure perfectly fits our needs. The R-PNN performs a sequential band-wise pansharpening using a single lightweight neural network, which is adapted on the fly to the target band, with no prior training but for the first band. This architecture allows us to address individually each band's needs. First, the desired level of spectral distortion is estimated based on the band features. Then, a band-adaptive optimization schedule is enacted to reach this target level, where the spatial loss is carefully balanced with the spectral loss and switched on and off according to a hysteresis-like schedule in order to avoid undesired local minima. In addition, the spatial loss itself is redefined to account for inverted PAN-HS correlations that can occur in specific spectral ranges. Experimental results on the benchmarking toolbox [40] prove the effectiveness of the proposed solution, which is competitive with the SotA on real full-resolution images in terms of global indices and much superior in terms of spectral quality consistency.

In summary, the proposed method inherits the appealing properties of the R-PNN and adds further strengths.

1) Fully unsupervised and zero-shot, with no need for prior training, no cross-dataset generalization issues.
2) Highly flexible: the number of spectral bands does not need to be fixed.
3) Uniform distortion along the whole spectral range, with target level defined by the user through a single and easily interpretable global parameter.
4) SotA performance on real full-resolution images, especially in terms of spectral quality.
5) Scalable complexity, which allows for tradeoff computation for quality based on application needs.

In the rest of this article, Section II provides some necessary background concepts. Section III describes the proposed solution. Section IV presents and comments on comparative experimental results. Section V draws conclusions and outlines future perspectives.

II. Background

The solution proposed in this article moves from our recent rolling PNN method [72], which we briefly summarize here, highlighting its distinctive features. Then, we analyze the issue of spectral fidelity for the current state-of-the-art methods.

A. Rolling PNN

Compared to the more familiar case of pansharpening of MS images, in the HS case, the generalization of DL models is usually more challenging for several reasons: less training data, richer spectral information, low correlation between PAN and spectral bands outside the visible spectrum, larger resolution ratios, and variability of the number of bands due to several technical reasons. Starting from the above observations and inspired by [37] and [69], we proposed the R-PNN [72], a CNN-based band-wise pansharpening solution where model parameters are not pretrained but generated on the fly on the same target image at inference time. Unlike previous works, suited to the MS case, in the R-PNN, the PAN is fused with only one band at a time, rather than all of them, and the tuned model is passed from one band to another for targeted adaptation. The overall band-wise pansharpening process is summarized in Fig. 1, with the main symbols used here and in the remainder of this article described in Table I.

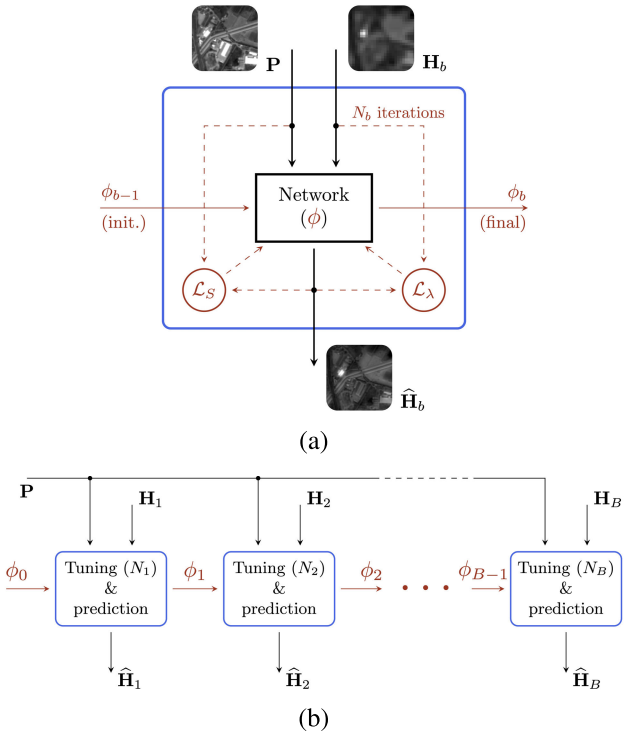


Fig. 1. R-PNN flowchart. Each module (a) inherits the initial weights, ϕ_{b-1}, from the previous module, fine-tunes them on the current band, and passes the updated version, ϕ_b, to the next module, as depicted in (b). The bth band is pansharpened with its own tuned parameters ϕ_b. N_b is the number of tuning iterations assigned to band b.

TABLE I
Main Symbols

Symbol	Dimensions	Meaning
w, h	Scalars	HS width and height, respectively
W, H	Scalars	PAN width and height, respectively
R	Scalar	Resolution ratio ($R = H/h = W/w$)
B	Scalar	# of HS bands
$\mathbf{H}$	$[w \times h \times B]$	Original HS image
$\mathbf{P}$	$[W \times H]$	Original PAN image
$\widetilde{\mathbf{H}}$	$[W \times H \times B]$	$R \times R$ upscaled version of $\mathbf{H}$
$\widehat{\mathbf{H}}$	$[W \times H \times B]$	Pansharpened version of $\mathbf{H}$
$\mathbf{D}$	$[W \times H \times B]$	Estimated details from $\mathbf{P}$ and $\widetilde{\mathbf{H}}$
$\mathbf{X}_b$	$[* \times *]$	b-th band of $\mathbf{X}$, $[* \times * \times B]$
$\mathcal{L}_\lambda$, $\mathcal{L}_S$, $\mathcal{L}$	Scalars	Spectral, spatial and overall loss functions

In mathematical terms, the network of Fig. 1(a) performs the mapping $\widehat{\mathbf{H}}_b = f(\mathbf{H}_b, \mathbf{P}; \phi)$, where ϕ is a set of trainable parameters. For each band, b (except band 1 starting from pretrained parameters), a tuning process starts from the initial parameters, $\phi^{(0)} = \phi_{b-1}$, inherited from the previous band, and runs for N_b iterations to get the tuned parameters $\phi_b = \phi^{(N_b)}$. N_b is proportional (up to a bounding value) to the spectral distance between bands b and $b-1$. In fact, HS datasets

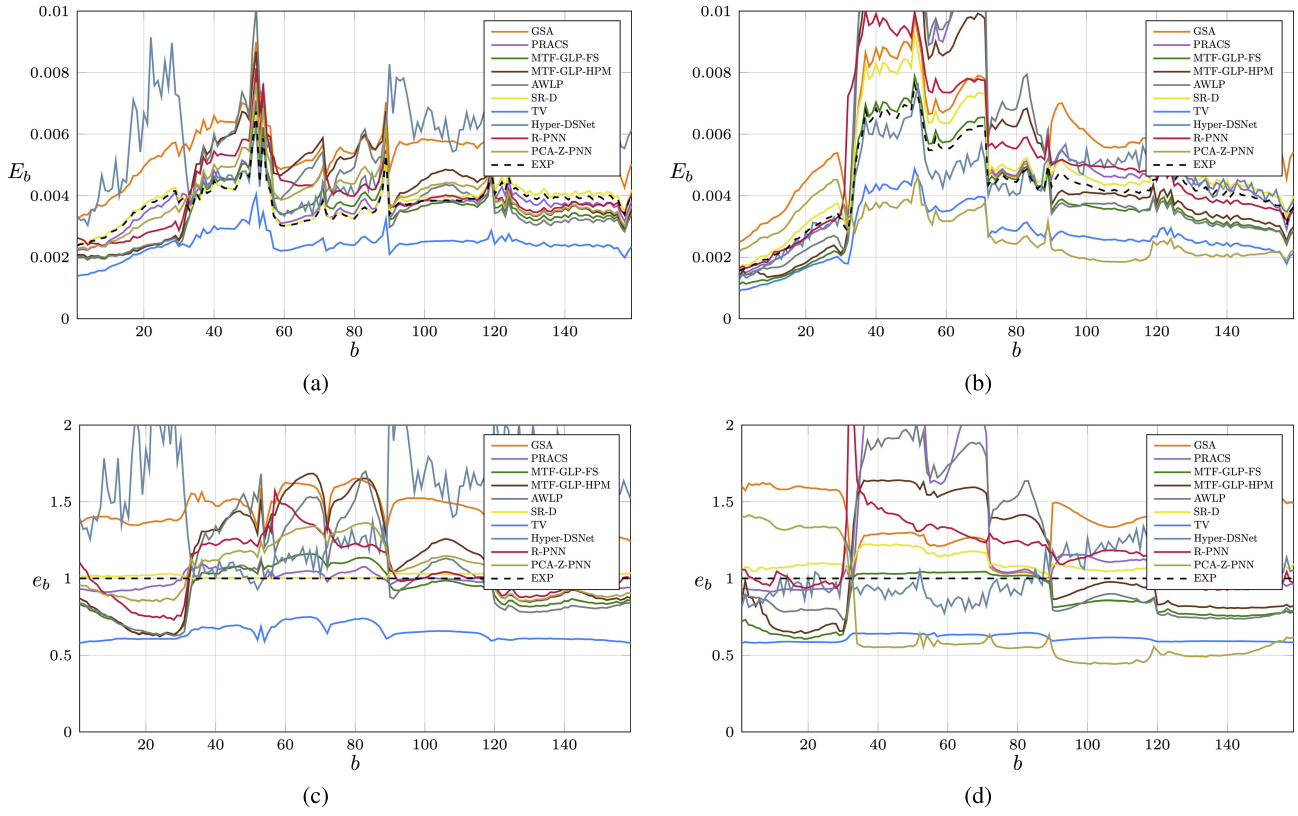


Fig. 2. Reprojection spectral error curves, E_b, for the PRISMA images: (a) Cagliari and (b) Udine, and (c) and (d) corresponding normalized curves e_b.

often present spectral gaps due to discarded bands, and longer training helps make up for reduced band similarity.

Each tuning iteration is carried out on the same components $\mathbf{H}_b$ and $\mathbf{P}$ to be fused, thanks to an unsupervised multitask loss which balances spectral and spatial consistencies of the fused image $\widehat{\mathbf{H}}_b$

$$\mathcal{L} = \mathcal{L}_\lambda\left(\widehat{\mathbf{H}}_b, \mathbf{H}_b\right) + \beta\mathcal{L}_S\left(\widehat{\mathbf{H}}_b, \mathbf{P}\right) \tag{1}$$

with β being a balancing hyperparameter. The spectral consistency is given by

$$\mathcal{L}_\lambda\left(\widehat{\mathbf{H}}_b, \mathbf{H}_b\right) = \left\|\widehat{\mathbf{H}}_b^{(\downarrow)} - \mathbf{H}_b\right\|_1 \tag{2}$$

where $\widehat{\mathbf{H}}_b^{(\downarrow)}$ is the downscaled (lowpass filtered and decimated) version of $\widehat{\mathbf{H}}_b$. The spatial consistency, instead, is given by the average *local* correlation between $\widehat{\mathbf{H}}_b$ and $\mathbf{P}$. Given the correlation coefficient $\rho(s)$, computed on a small $\sigma \times \sigma$ neighborhood of pixel s, the loss is defined as

$$\mathcal{L}_S\left(\widehat{\mathbf{H}}_b, \mathbf{P}\right) = \left\langle \left[\rho^{\max}(s) - \rho(s)\right]_+ \right\rangle \tag{3}$$

where $\langle x \rangle$ indicates spatial average, $[x]_+ = \max(0, x)$, and $\rho^{\max}(s)$ is a rough estimate of the expected local correlation, computed on reduced resolution data. Therefore, this spatial loss aims at maximizing $\rho(s)$, but only as long as it does not exceed a reasonable target value, $\rho^{\max}(s)$.

B. Analysis of Spectral Distortion

For all methods of the benchmarking toolbox [40], and for two different PRISMA test images, we computed the band-wise reprojection error[2]

$$E_b = \left\|\widehat{\mathbf{H}}_b^{(\downarrow)} - \mathbf{H}_b\right\|_1 \tag{4}$$

obtained by comparing the downscaled (smoothed and decimated) version of the pansharpened image with the input HS image. This is the same distance used in the R-PNN (2), but other metrics provide similar results. In Fig. 2(a) and (b), we plot results only for the methods characterized by the best full-resolution spectral quality.

The error is clearly band-dependent. Almost all methods follow a common pattern, certainly related also to the different levels and properties of noise in different bands. Even the simple EXP interpolator, which is a 23-tap polynomial approximation of the ideal interpolator, follows the same dynamics. The average level of each curve provides information about the global distortion incurred by the method (remember that we are neglecting spatial quality in this analysis) but the detailed behavior of the curves shows that numbers vary significantly from band to band, and that the average distortion may be a poor indicator of the actual distortion observed in certain frequency ranges. Under this point of view, a uniform level of distortion would certainly be an appreciable feature.

To simplify the analysis of results, let us consider a normalized version of the same data. Since the EXP interpolator works exclusively on $\mathbf{H}$, without adding any high-frequency spatial detail coming from the PAN, it can be regarded as a

[2]Although this quantity is often referred to as spectral distortion, here we prefer to recall the presence of a downscaling step using the term reprojection.

method-independent "spectral reference." Therefore, we use its reprojection error E_b^{EXP} to relativize the profiles of Fig. 2 by defining a normalized error e_b as

$$e_b = E_b \,/\, E_b^{\text{EXP}}. \tag{5}$$

The resulting curves are plotted in Fig. 2(c) and (d). Spectral error profiles are now easier to analyze. As a "virtuous" example, we can mention TV, which, for both test images, keeps the normalized reprojection error low (about 60% of EXP) and almost constant over the whole spectral range. In contrast, other methods show significant imbalances between different spectral ranges. This may be partly due to spatial detail injection rules that do not properly account for the spectral-spatial interdependence. For example, several methods achieve very low spectral distortions in the visible range (first ∼30 bands) with a sharp performance impairment beyond the near infrared limit, possibly due to the inappropriate injection of spatial details in a range where the correlation between PAN and HS bands decreases. On the opposite side, we mention the case of PCA-Z-PNN, which follows a *divide-and-conquer* approach where the HS bands are split into two groups and separately processed. This split is clearly reflected in the almost piecewise uniform e_b curve [see Fig. 2(d)], with a relatively large distortion in the visible range and a much better behavior beyond that range.

Overall, this preliminary analysis motivates us to work on a pansharpening method that adapts to the different characteristics exhibited by HS bands in different spectral ranges and provides nearly uniform quality across the spectrum. The R-PNN, with its band-wise optimization procedure, is an excellent candidate to achieve this goal.

III. Proposed Method

A. Correlation-Based Spatial Loss for HS Images

In our proposal, we keep the two-component loss formulation (1) previously adopted [70] in unsupervised pansharpening and use the same spectral component of [72], based on the ℓ_1-norm (2), which shows good convergence properties and outperforms other norms, such as ℓ_2 or ERGAS, in terms of perceptual quality. In any case, the exact form of the spectral loss is not really critical as it is always obtained by comparing the pansharpened product $\widehat{\mathbf{H}}$ with a clear reference, the interpolated HS image $\widetilde{\mathbf{H}}$ or (after decimation) the low-resolution HS itself. We focus instead on the spatial loss term, which is more elusive as it is intimately related to the problem of assessing spatial quality without a reference. In the R-PNN, we used the correlation-based loss (3) originally proposed in [69] for MS pansharpening. The idea is to push the pansharpened band to have a high local correlation with the PAN, so as to exhibit the same spatial layout. Indeed, the correlation coefficient measures how well one image can be linearly predicted from another. Since pansharpened bands and PAN share the same spatial structure, they can be expected to be strongly correlated. On the other hand, each band has unique spectral properties that must be carefully preserved. Therefore, to prevent the injection of alien details from the PAN into the spectral band, the correlation itself, ρ, is capped by its estimate, $\rho^{\max}(s)$, computed on reduced resolution data.

This correlation-based loss proved extremely effective for MS pansharpening, ensuring results characterized by very high spatial fidelity [69], [70]. Its extension to the HS case, however, is more controversial. In fact, the basic assumption is that $\rho(s)$ is generally high and certainly positive. While this is more than reasonable for the MS case, it becomes largely questionable in the HS case because many spectral bands fall outside the visible range (i.e., the PAN range) and exhibit a lower correlation with it, and sometimes the so-called inversion phenomenon [74] with regions having strong but negative correlation.

To clarify this point, we carried out a correlation analysis on a sample PRISMA image (Kansas). Lacking the ideal pansharpened datacube, we estimate the local correlation on reduced resolution data, using the canonical resolution downgrade of the PAN to align it with the available HS datacube. We select two sample spectral bands which represent opposite behaviors, one in the visible spectrum, thus highly correlated with the PAN, the other outside this range and with much lower correlation. In Fig. 3 (top), we show the downgraded PAN $\mathbf{P}^{(\downarrow)}$, the two spectral bands, $\mathbf{H}_v$ (from the visible spectrum) and $\mathbf{H}_o$ (outside), together with the corresponding correlation maps, $\rho_v = \rho_{\mathbf{P}^{(\downarrow)}\mathbf{H}_v}$ and $\rho_o = \rho_{\mathbf{P}^{(\downarrow)}\mathbf{H}_o}$, respectively. The rightmost image is a quantized representation of ρ_o used to simplify its interpretation. At the bottom of the figure, the same items are shown for a meaningful close-up of the image (green box).

A few observations are in order. The sample band from the visible spectrum shows a very high correlation with the PAN, 0.95 on average, and never negative. This is a common behavior in the visible spectrum, more pronounced in the green range, and a little less toward the coastal and red/infrared limits. In this situation, maximizing the local correlation, without any upper bound, makes perfect sense. Outside the visible spectrum, however, the correlation field has a much higher dynamics, and our sample band has an average correlation of just 0.33 with the PAN. Such a low value, however, does not signal a lack of dependence. The spatial layouts keep being mostly the same, as easily appreciated in the close-up image, but several regions exhibit a strong *negative* correlation. This is the case, for example, of the circularly shaped elements clearly visible in the close-up, which are very dark in the PAN, and very bright in $\mathbf{H}_o$. In addition, intermediate correlation values, between -0.6 and 0.6, occur at the boundaries between positively and negatively correlated regions just because of the finite size of the estimation window (6×6 in our experiment). This latter problem also impacts the estimation of the bounding map, $\rho^{\max}$, further reducing its effective resolution and making it scarcely reliable. An algorithm that disregards these phenomena may generate visible pansharpening artifacts, such as unnatural color variations and exaggerated contrasts. Moreover, conflicts may arise between the spectral and spatial terms of the loss, hindering the training or fine-tuning phases.

Based on the above observations, we decided to define a spatial loss term that maximizes the *absolute value* of the correlation with the PAN, to promote structural similarity

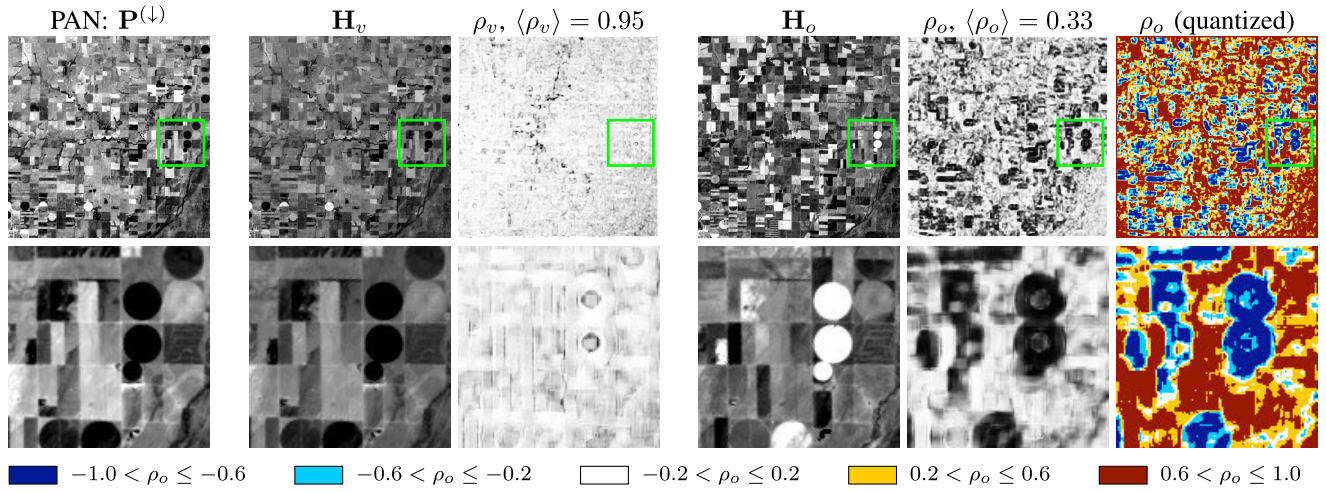


Fig. 3. PAN-HS local correlation analysis. (Top, from left to right) PAN image shifted at the HS spatial resolution ($\mathbf{P}^{(\downarrow)}$), two sample HS bands paired with the corresponding map of local correlation with the PAN, from visible ($\mathbf{H}_v$, ρ_v) and NIR-SWIR ($\mathbf{H}_o$, ρ_o) spectral ranges, and (rightmost) the quantized version of the correlation map ρ_o. (Bottom) Close-up on a meaningful detail (highlighted with a green box).

irrespective of the correlation polarity

$$\mathcal{L}_S\left(\widehat{\mathbf{H}}_b, \mathbf{P}\right) = \langle 1 - |\rho(s)| \rangle. \tag{6}$$

Note that we have also removed the bounding map $\rho^{\max}$ due to its limited reliability. In fact, a suitable hysteresis-based optimization schedule will prevent the injection of inappropriate PAN details in the pansharpened image.

Armed with our new loss, comprising the original spectral term (2) and the new spatial term (6), we can now pursue our original goal, that is, devising a pansharpening algorithm that provides a uniform quality across the whole spectrum. More precisely, considering that the various bands experience very different noise intensities, we aim at achieving a uniform *relative* quality, measured by the error normalized with respect to the EXP ideal interpolator (5).

B. Network Architecture

As said before, we rely on rolling pansharpening [72] as it allows us to operate band-wise. With this strategy, only one low-resolution interpolated spectral band is given in input, together with the PAN, and only one high-resolution spectral band is provided in output. Therefore, any pansharpening network can be adapted to the purpose by setting appropriately the depth of the input and output layers. The network must be trained from scratch for the first band and then refined for each new band by target-adaptive tuning. Of course, for an HS image, this latter process must be repeated many times, which calls for the intrinsic requisite of a lightweight network.

We have implemented and tested several low-complexity solutions from the MS pansharpening literature and verified experimentally that network structure and capacity have a minor impact on performance provided some minimal features are guaranteed: 1) no less than three convolutional layers; 2) a global receptive field of a reasonable size; and 3) use of a global skip connection (residual net). Given the above indications, we decided to simply keep the network used in the R-PNN, a single-band residual version of the PNN [37], which meets all these criteria. Table II summarizes the network layers and connections.

TABLE II
NETWORK LAYERS AND TOPOLOGY

Layer	Size	Kernel	Outputs
Input	$W \times H \times 2$	-	$(\widetilde{\mathbf{H}}_b, \mathbf{P})$
Conv+ReLU	$W \times H \times 48$	7×7	-
Conv+ReLU	$W \times H \times 32$	7×7	-
Conv	$W \times H \times 1$	5×5	$\mathbf{D}_b$
Output (add gate)	$W \times H \times 1$	-	$\widetilde{\mathbf{H}}_b + \mathbf{D}_b$

C. Hysteresis-Based Training Procedure

Once the network architecture is fixed, we can focus on optimizing network training and fine-tuning. To this end, we can play with several hyperparameters, first of all the learning rate α, the weight β that balances the spectral and spatial terms in the loss, and the number of iterations N_b. To gain insights into the problem, we carried out some preliminary experiments using various (α, β) pairs and studying their impact on the two loss components, $\mathcal{L}_S$ and $\mathcal{L}_\lambda$ (these losses depend also on the band, b, but we avoid a further subscript for the sake of clarity). To avoid initialization issues, we trained the network from scratch and used a very large number of iterations, $N_b = 1000$. In Fig. 4, for two different bands, one from the visible range (top) and one outside it (bottom), we plot the evolution of the loss components in the $(\mathcal{L}_{\lambda/E}, \mathcal{L}_S)$ plane as a function of the number of iterations. For the reasons explained before, for the spectral loss we consider the normalized value $\mathcal{L}_{\lambda/E} = \mathcal{L}_\lambda / \mathcal{L}_\lambda^{\text{EXP}}$. Each curve starts from the top-right point and evolves over time toward lower values of the losses.

For the "visible" band, everything works quite well. At the end of the process, $\mathcal{L}_S$ is always very low, which is to be expected given the very high correlation with the PAN, and also the spectral quality is quite good, with $\mathcal{L}_{\lambda/E}$ going largely below 1 for some (α, β) pairs, thus improving significantly over the reference value provided by the EXP method. For the "outside" band, the story is completely different. $\mathcal{L}_S$ never

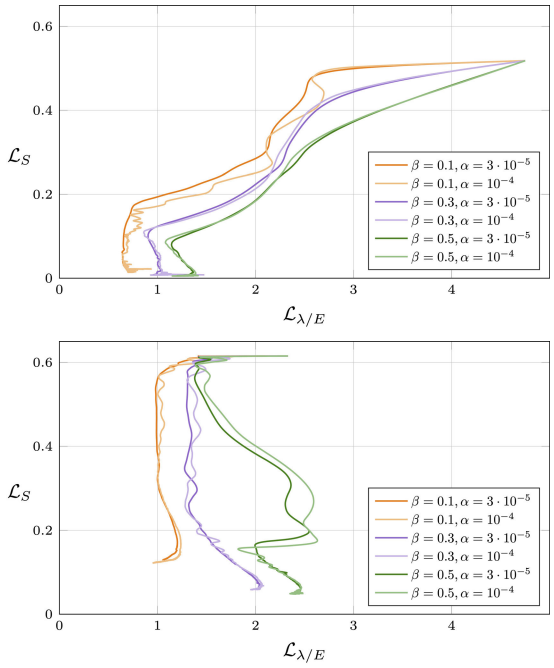


Fig. 4. Loss trajectories in the $(\mathcal{L}_{\lambda/E}, \mathcal{L}_S)$ plane when tuning models with various choices of the α and β parameters. Tuning starts from random initial weights and proceeds for a large number (1000) of iterations. (Top) Sample bands from the visible range and (bottom) outside it.

approaches 0, and this was expected considering the much lower correlation of this band with the PAN. However, also $\mathcal{L}_\lambda$ remains rather large and only in some cases it approaches the reference $\mathcal{L}_\lambda^{\text{EXP}}$ level. The loss trajectories are especially informative. In all cases, $\mathcal{L}_S$ improves regularly as the process goes on, but this seems to affect negatively $\mathcal{L}_\lambda$, which, after an initial improvement, impairs markedly and never recovers. Contrary to what happens for the visible band, $\mathcal{L}_{\lambda/E}$ never goes below 1.

This behavior gave us the idea to use a more sophisticated training schedule, where the balance between spectral and spatial loss components is adjusted dynamically during the process. In particular, we decided to set $\beta = 0$ at the beginning and fine-tune the network based only on the spectral loss, so as to bring it down to a desired level, say $\mathcal{L}_{\lambda/E} = \gamma_{\text{LOW}}$. Then, we turn the $\mathcal{L}_S$ loss component back on, but always keeping the spectral quality under control. In fact, as soon as $\mathcal{L}_{\lambda/E}$ exceeds a second higher threshold, γ_{HIGH}, the spatial loss component is turned off again, and $\mathcal{L}_{\lambda/E}$ is brought again to the lower level. Our intuition was that this optimization cycle could bring the curve to a new point in the trajectory, with the same spectral loss as before but a lower spatial loss. Fig. 5 shows that this is actually the case. In this figure, we plot the same curves as before but organized differently. Each chart now refers to a different (α, β) pair and compares the $(\mathcal{L}_{\lambda/E}, \mathcal{L}_S)$ trajectories obtained with flat training and with the proposed on–off schedule. For the visible band, the proposed schedule does not really improve over the baseline (the mechanism could even be switched off). On the contrary, for the outside band, the gain is striking. The desired level of spectral distortion is readily achieved with all (α, β) pairs. Then $\mathcal{L}_S$ is gradually reduced while $\mathcal{L}_\lambda$ keeps oscillating in the desired range. The final value of $\mathcal{L}_S$ is somewhat larger than with flat training, but this is quite reasonable for an "outside" band, intrinsically less correlated with the PAN, and does not correspond to a worse spatial quality. A further interesting property emerging from this analysis is the remarkable stability of the trajectories, almost identical to one another as the hyperparameters change, which suggests a good robustness with respect to changing sources, namely, a good generalization ability. Finally, note that most of the final iterations are used to gain only minor improvements in $\mathcal{L}_S$, and tuning could be safely stopped much earlier. This last observation allowed us to define a computationally efficient stopping rule. For the current band, b, we count both the total number of iterations N_b and the number of iterations with active spatial loss ($\beta > 0$), $N_{b|S}$. Tuning stops when either one exceeds a predefined limit, whose setting is discussed in Section IV. Note that also the value of β to be used when the spatial loss is active is not fixed once and for all, but only after a suitable warm-up. A large value, β_0, is used initially, but this is repeatedly halved if it causes an excessive increase in spectral loss, down to a minimum of $\beta_0/8$. This mechanism relieves us from the burden of selecting a correct value for β, which, on the other hand, can change from band to band. Even so, a few parameters remain to be set, a problem that will be analyzed in some detail in Section IV.

Fig. 6 illustrates an example of tuning with hysteresis for a few bands of a sample image. (a) and (b) Evolution of the (normalized) spectral and, respectively, spatial loss components as the tuning proceeds. Red lines highlight ON phases, where both loss terms are active. In this example, we have fixed $\gamma_{\text{LOW}} = 0.72$ and $\gamma_{\text{HIGH}} = 0.8$. Several interesting observations can be made.

1) Most of the time, the model optimized for the previous band works very well on the new one, for example, 55→56 or 58→59. In some cases, for example, 56→57, the transfer is less effective, which significantly impacts on quality and calls for a tuning overhead to bring the spectral loss within the hysteresis limits. This latter case, however, occurs rarely, typically when going from the last band of a correlation block to the first band of another one.
2) Different average levels of spatial loss are observed for different bands, reflecting their different relationship with the PAN.
3) Boosting sharpness ($\mathcal{L}_S$ ON) causes an increase of the spectral loss $\mathcal{L}_\lambda$ and the need for multiple hysteresis cycles to keep it within the desired limits.

Given the oscillatory behavior of both losses and some overshooting due to momentum, the last model provided by the fine-tuning procedure is not necessarily the best-performing one. Therefore, in our implementation, we keep memory of all of them and output the model with the lowest spatial loss among those that satisfy the constraints on the spectral loss.

Algorithm 1 describes the proposed method more formally using a high-level pseudo-code.

IV. Experimental Analysis

In this section, we perform the comparative performance assessment of the proposed method. First, we set some key

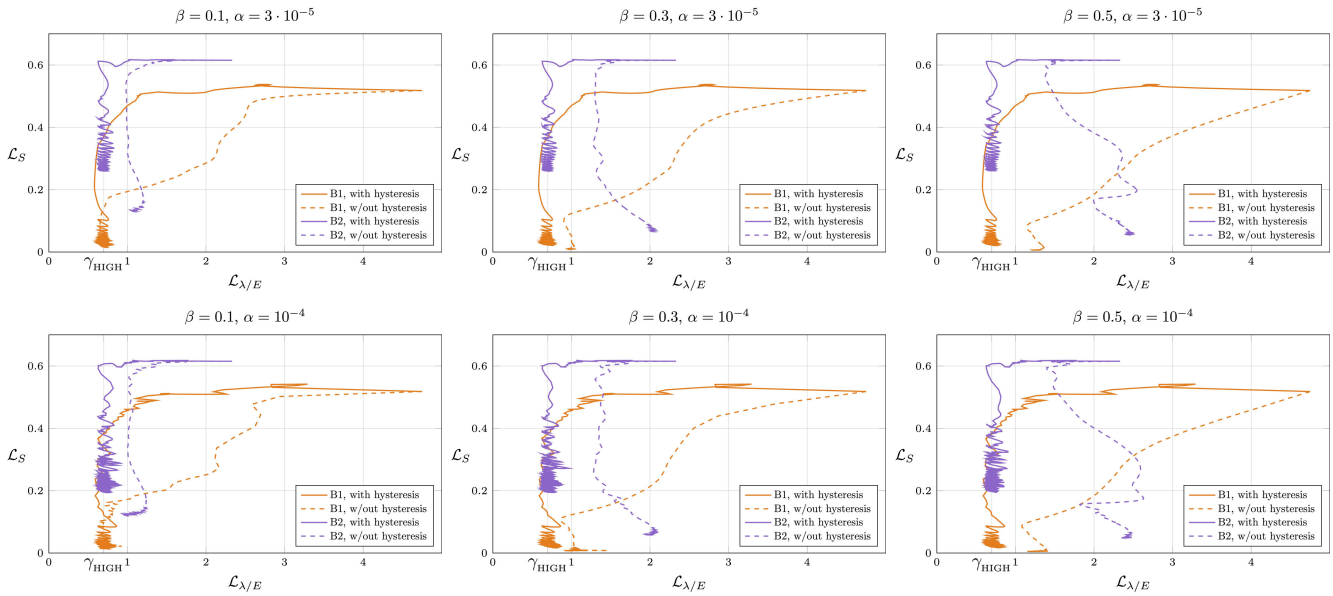


Fig. 5. Loss trajectories in the $(\mathcal{L}_{\lambda/E}, \mathcal{L}_S)$ plane when tuning models with various choices of the α and β parameters. Tuning starts from random initial weights and proceeds for a large number (1000) of iterations. In each chart, the curves refer to two different bands, B1 from the visible spectrum (violet) and B2 outside it (orange), and two different tuning schedules, the proposed hysteresis-like strategy (solid lines) and flat tuning (dashed lines).

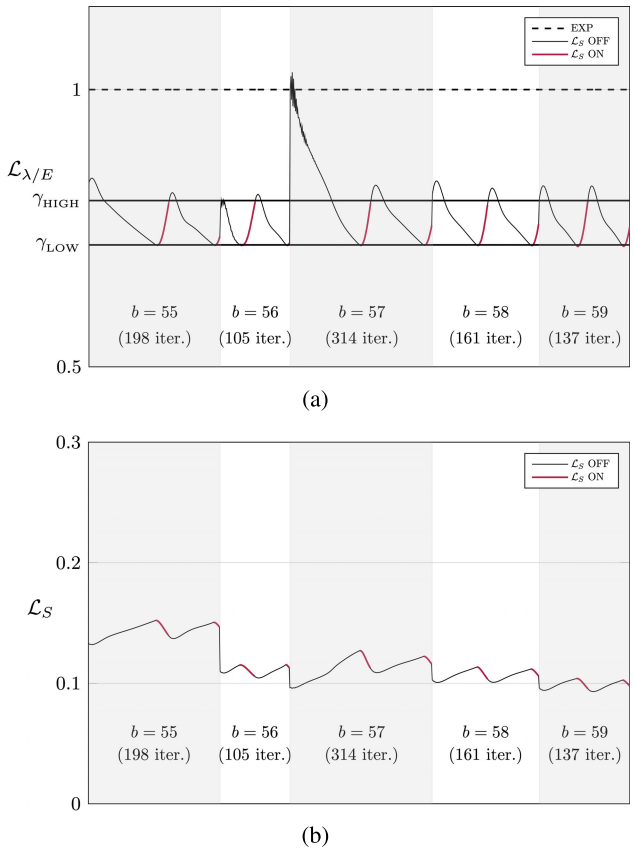


Fig. 6. Close-up (bands 55–59) of the evolution of (a) spectral and (b) spatial loss terms during the hysteresis-based tuning with model transfer.

Algorithm 1 ρ-PNN Pseudo-Code

Input: $\mathbf{P}$, $\mathbf{H}$, net (network structure)
Hyperparameters: γ_{HIGH}, γ_{LOW}, β_0, η, N_0, $N_S^{\max}$, ϵ
Output: $\widehat{\mathbf{H}}$

$B \leftarrow$ size($\mathbf{H}$, 3) ▷ # of bands
$\mathbf{c} \leftarrow$ correlations($\mathbf{H}$) ▷ $\mathbf{c}_b = \text{corrcoef}(\mathbf{H}_b, \mathbf{H}_{b-1})$; $\mathbf{c}_1 = 0$
$\phi \leftarrow$ weightInit(net) ▷ net weights init.
for $b \leftarrow 1 : B$ **do**
  $N_b^{\max} \leftarrow$ iterationUpperBound($\mathbf{c}$, N_0, B, η) ▷ Eqs. (8), (10)
  $\widetilde{\mathbf{H}}_b \leftarrow$ interpolator($\mathbf{H}_b$)
  $\mathcal{L}_\lambda^{\text{EXP}} \leftarrow$ getSpectralLoss($\widetilde{\mathbf{H}}_b$, $\mathbf{H}_b$)
  $N_b, N_{b|S} \leftarrow 0$ ▷ counter init.
  $\beta \leftarrow$ warmUp($\mathbf{H}_b$, $\mathbf{P}$, net, ϕ, β_0, ϵ, α) ▷ β warm-up
  ON $\leftarrow$ false ▷ $\mathcal{L}_S$ flag init.
  while $\{N_{b|S} < N_S^{\max}\}$ **and** $\{N_b < N_b^{\max}\}$ **do** ▷ Eq. (7)
    $\phi, \mathcal{L}_\lambda, \mathcal{L}_S \leftarrow$ optimizationStep($\mathbf{H}_b$, $\mathbf{P}$, net, ϕ, β, ON)
    $N_b \leftarrow N_b + 1$
    if ON **then**
      $N_{b|S} \leftarrow N_{b|S} + 1$
    end if
    if $\mathcal{L}_{\lambda/E} > \gamma_{\text{HIGH}}$ **then**
      ON $\leftarrow$ false ▷ switch-off
    else if $\mathcal{L}_{\lambda/E} < \gamma_{\text{LOW}}$ **then**
      ON $\leftarrow$ true ▷ switch-on
    end if
  end while
  $\widehat{\mathbf{H}}_b \leftarrow$ predict($\mathbf{H}_b$, $\mathbf{P}$, net, ϕ) ▷ inference on band b
end for
return $\widehat{\mathbf{H}}$

parameters of the method based on preliminary experiments on the validation set of the benchmarking toolbox [40]. Then, after discussing the assessment methodology, we carry out an objective performance analysis on reduced-resolution and full-resolution test data of the same toolbox, and finally perform a visual analysis of the results.

A. Parameter Setting

1) Setting the Target for Spectral Loss: A key question for our approach is: to what extent can the spectral distortion be reduced without causing appreciable spatial distortion? To answer this question, we carried out a set of experiments on five bands {B1, ..., B5} of the validation dataset, sampled uniformly from the whole spectral range. For each band,

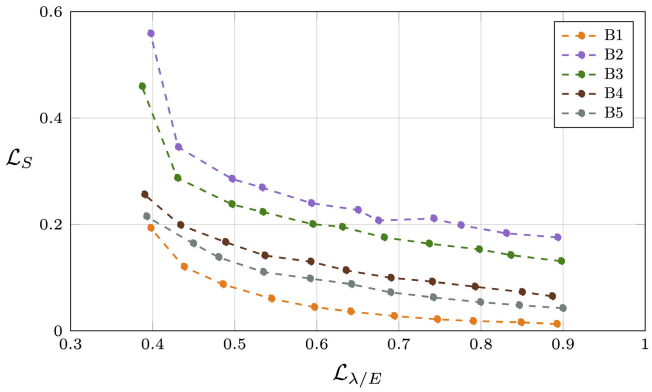


Fig. 7. Loss points in the $(\mathcal{L}_{\lambda/E}, \mathcal{L}_S)$-plane for various bands after thorough hysteresis-based tuning as a function of the γ_{HIGH} and γ_{LOW} parameters.

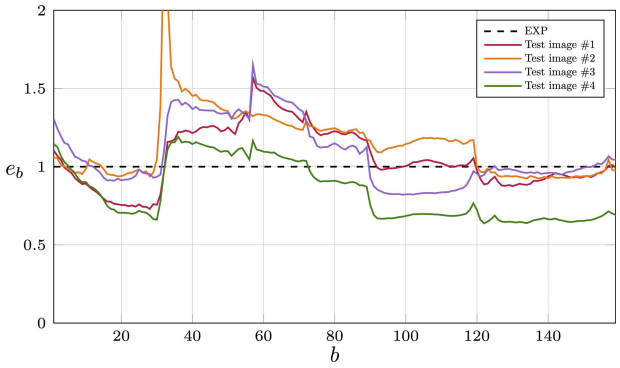


Fig. 8. R-PNN normalized reprojection error over four PRISMA images.

we trained the model from scratch (no cross-band model transfer) with our hysteresis-based schedule, for virtually infinite iterations, using always $\alpha = 10^{-5}$ and $\beta = 0.5$, but varying γ_{HIGH} in $\{0.40, 0.45, \ldots, 0.90\}$, with $\gamma_{\text{LOW}} = 0.9\gamma_{\text{HIGH}}$.

In Fig. 7, we show the final loss values achieved as points in the $(\mathcal{L}_{\lambda/E}, \mathcal{L}_S)$ plane. For all bands, we observe a similar behavior. The lowest spatial loss is obtained for $\gamma_{\text{HIGH}} = 0.9$ and differs from band to band depending on its noise level and especially its similarity with the PAN. For bands in the visible spectrum, for example, B1, with very high local correlation with the PAN, $\mathcal{L}_S$ goes almost to 0 (0.03), while it is much higher for other bands, for example, B2, from the near-IR range, with $\mathcal{L}_S$ close to 0.2. In all cases, the spatial loss increases as γ_{HIGH} decreases, as expected, but just slowly at the beginning, and a steep increase is observed only when $\mathcal{L}_{\lambda/E}$ falls below 0.5, that is, $\mathcal{L}_\lambda$ is 50% of the reference $\mathcal{L}_\lambda^{\text{EXP}}$ value. Based on this consistent behavior, we decided to set $\gamma_{\text{HIGH}} = 0.65$ and $\gamma_{\text{LOW}} = 0.59$ in the upcoming experiments.

2) Limiting the Number of Iterations: Besides ensuring high and consistent quality, we want to keep complexity under control by limiting the number of tuning iterations. However, as also shown by Fig. 6, the previous-band model is not always a good starting point, in which case many iterations are required to bring the loss down to acceptable values. This appears clearly in Fig. 8, which shows the normalized reprojection error of R-PNN over four PRISMA images. The error spikes over the first band of a correlation block, then

TABLE III
MAIN HYPERPARAMETERS OF THE PROPOSED METHOD

Param	Value	Meaning
γ_{HIGH}	0.65	Upper limit for $\mathcal{L}_{\lambda/E}$
γ_{LOW}	0.59	Lower limit for $\mathcal{L}_{\lambda/E}$
β_0	2	Initial β, weight of the spatial loss term
ϵ	0.007	Growth factor of $\mathcal{L}_\lambda$ for β adaptation
$N_S^{\max}$	20	Target # of tuning iter.s with active spatial loss
N_0	80	Max # of tuning iter.s when $\mathbf{c}_b = 1$
η	30	Global param. controlling total # of tuning iter.s

stabilizes at lower values in subsequent bands due to the high interband similarity. Based on this observation, we decided to stop tuning when either one of two conditions holds:

$$\left\{N_{b|S} = N_{b|S}^{\max}\right\} \text{ OR } \left\{N_b = N_b^{\max}\right\}. \tag{7}$$

Most of the time, the spectral loss quickly reaches the desired value, and most iterations also involve the spatial loss. In these cases, the active constraint is on $N_{b|S}$, and we set the maximum to a fixed value for all bands, $N_{b|S}^{\max} = N_S^{\max}$. More rarely, when subsequent bands are weakly correlated, reducing the spectral loss is more difficult, and a limit on the total number of iterations $N_b^{\max}$ becomes necessary to prevent an inordinate number of them. This latter limit, however, must be carefully set to make sure that a reasonable number of iterations involve also the spatial loss. Therefore, we decided to make this limit depend on interband correlation, that is,

$$N_b^{\max} = N_0 + \Delta N(1 - \mathbf{c}_b) \tag{8}$$

with c_b being the correlation coefficient between bands b and $b-1$. Therefore, the maximum number of iterations is $N_0 \geq N_S^{\max}$ in the case of perfect correlation ($\mathbf{c}_b = 1$) and $N_0 + \Delta N$ in the case of orthogonal bands ($\mathbf{c}_b = 0$). With this rule, we can easily set an overall bound to the total number of iterations N as follows:

$$N = \sum_b N_b \leq N^{\max} = \sum_b N_b^{\max} = (1+\eta)BN_0 \tag{9}$$

with η being a suitable overhead multiplier to be set by the user and B being the number of bands. Therefore, one can set $N^{\max}$ in advance based on complexity issues, and N_0 and η accordingly. Then, ΔN will be given by the (easy to prove) formula

$$\Delta N = \frac{\eta B N_0}{B - \sum_b \mathbf{c}_b}. \tag{10}$$

Fig. 9 shows an example of how iterations distribute over the various bands of a PRISMA image. Here, we set $N_S^{\max} = 20$, $N_0 = 50$, and $\eta = 8$. $N_{b|S}$ is shown in yellow and N_b in brown, with its upper bound $N_b^{\max}$ in light gray. Although a maximum of $(1+\eta)N_0 = 450$ iterations per band is allowed on average, only 47 are used on average. Most of the times, tuning stops because $N_{b|S}$ reaches its maximum, $N_S^{\max} = 20$. The total number of iterations rarely exceeds $N_0 = 50$ and only for a group of bands, on the boundary between the visible and NIR ranges, the $N_b = N_b^{\max}$ exit condition is activated.

Table III summarizes the most relevant hyperparameters of the proposed method, their selected values, and their meaning.

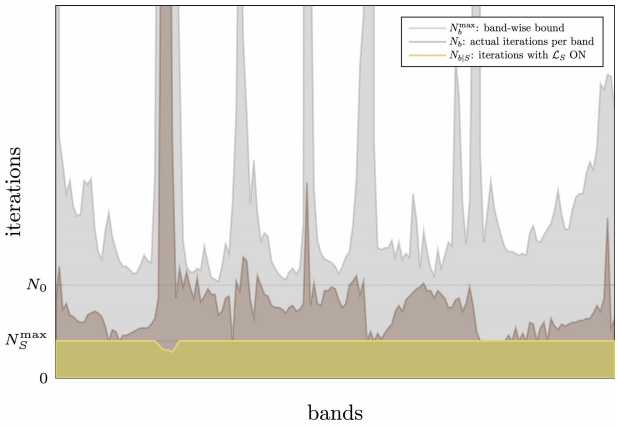


Fig. 9. Distribution of tuning iterations for a test PRISMA image. On average, 47 iterations/band are used, almost always 20 of them with spatial loss ON.

TABLE IV
PAN–MS PRISMA BENCHMARK DATASETS FOR TESTING

Date	Location	Land covers
05/09/22	Cagliari, Italy	Roofs, Streets, Crops, Hills, Water
24/08/23	Udine, Italy	Roofs, Streets, Crops, Water
08/09/23	Ford County, Kansas	Streets, Crops
20/11/23	Macuspana, Tabasco	Streets, Crops
RR test image size: 600×600 (from a 3600×3600 real FR image)		
Real FR test image size: 1200×1200		
Resolution ratio $R = 6$		
Bands $B = 159/239$		
Resolution: 30 m (RR) or 5 m (FR)		

B. Assessment Methodology

To assess the quality of the proposed solution, we take advantage of the benchmarking toolbox [40] which provides a publicly available set of rich and diverse PAN-HS PRISMA images, a large set of SotA comparative methods, and several performance evaluation tools based on the most credited assessment indices.

PRISMA satellites are equipped with a sensor operating in the 400–700-nm spectral range that provides a PAN image with 5-m spatial resolution, and two HS sensors that acquire 66 bands at VNIR wavelengths, 400–1010 nm, and 173 bands at SWIR wavelengths, 920–2505 nm, with 30-m resolution. The dataset released by [40] includes 16 images (12 for training and validation, and 4 for testing) comprising 159 spectral bands after removing noisy and uninformative channels. For our experiments, we use the four test images of the Toolbox (see Table IV). From each original PRISMA PAN-HS pair (PAN/HS size: 3600 × 3600/600 × 600), two test datasets were created: an FR one (PAN/HS size: 1200 × 1200/200 × 200) obtained by cropping; an RR one (PAN/HS/GT size 600 × 600/100 × 100/600 × 600) obtained by resolution downgrading of the whole PAN-HS pair, with the original FR HS playing the role of GT.

Table V gathers the comparative methods which can be roughly grouped in four categories: component substitution (CS), MRA, model-based optimization (MBO), and DL. The interested reader is referred to [40] for details on these methods.

TABLE V
HS PANSHARPENING METHODS

Name	Ref	Summary
EXP		Approximation of the ideal interpolator
Component Substitution (CS) methods		
GSA	[24]	Gram-Schmidt adaptive component substitution
BT-H	[27]	Brovey transform with haze correction
BDSD-PC	[75]	Band-dependent spatial detail injection with physical constraint
PRACS	[26]	Partial replacement adaptive CS
Multiresolution Analysis (MRA) methods		
MTF-GLP-FS	[75]	Modulation Transfer Function (MTF)-matched Generalized Laplacian Pyramid (MTF-GLP) with fusion rule at full scale
MTF-GLP-HPM	[77]	MTF-GLP with high pass modulation
MTF-GLP-HPM-R	[78]	MTF-GLP-HPM with regression-based spectral matching
AWLP	[28]	Additive wavelet luminance proportional
MF	[30]	Nonlinear decomposition with morphological filters
Model-Based Optimization (MBO) methods		
HySURE	[44]	Bayesian estimation with vector total variation prior
SR-D	[79]	Sparse representations-based detail injection
TV	[31]	Total variation-based pansharpening
Deep Learning (DL) methods		
HyperPNN	[57]	7-layer net with spectral encoder-decoder structure
HSpeNet	[58]	Advanced version of HyperPNN, with deeper architecture and Spectral Angle Mapper (SAM) loss term
DHP-DARN	[59]	Deep residual channel-spatial attention net with Deep Image Prior (DIP) for HS resizing
DIP-HyperKite	[62]	Overcomplete net with DIP for HS resizing
HyperDSNet	[80]	Spectral attention-based detail injection from deep-shallow features
R-PNN (unsup.)	[72]	Band-wise pansharpening using modified Z-PNN [70] with tuning propagation
PCA-Z-PNN (unsup.)	[73]	Z-PNN model with PCA-based input reduction

TABLE VI
HS PANSHARPENING QUALITY ASSESSMENT INDEXES

RR assessment	
ERGAS	*Erreur Relative Globale Adimensionnelle de Synthése* [83]
SAM	Spectral Angle Mapper [84]
$Q2^n$	Multiband extension [86] of Universal Image Quality Index [86]
FR assessment	
D_λ	Khan's spectral distortion index [87], [88]
D_S	Spatial distortion index [87], [89]
D_ρ	Correlation-based spatial distortion index [82]

The assessment follows the well-established Wald's protocol [81] which considers two contexts: a reduced resolution framework, where suitably degraded (in resolution) datasets are used for assessment *with* GT (where the ground truth is given by the original full-resolution (FR) HS image), and an FR framework for assessment *without* GT but using the original data as input. Accordingly, the quality indices, gathered in Table VI, are split into two groups for reduced resolution (RR) and FR assessment.

As is well known, the assessment of spatial quality in the FR context (hence, without a GT) is still an open and challenging problem. Therefore, to obtain a more comprehensive evaluation of performance, besides the D_S index used in the

TABLE VII
RESULTS AT REDUCED RESOLUTION. BEST SCORE IS IN BOLD GREEN, TOP-5 IN GREEN, AND WORST-5 IN RED

Method	ERGAS					SAM					$Q2^n$				
	Cagliari	Udine	Ford	Tabasco	Avg.	Cagliari	Udine	Ford	Tabasco	Avg.	Cagliari	Udine	Ford	Tabasco	Avg.
(Ideal)	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1
EXP	1.7716	3.9587	1.6921	4.3650	2.9468	2.3073	4.6288	2.8528	6.3524	4.0353	0.5971	0.6453	0.7984	0.5935	0.6586
GSA	0.9389	2.4243	1.0008	2.8065	1.7926	1.8191	3.5730	2.2570	5.7455	3.3486	0.8739	0.8110	0.9201	0.7955	0.8501
BT-H	1.2784	3.4898	1.2462	3.1015	2.2790	2.3857	4.6059	2.5563	6.3532	3.9753	0.8180	0.7166	0.8941	0.7837	0.8031
BDSD-PC	1.1664	2.7627	1.2637	3.9316	2.2811	1.9439	5.0116	2.6646	7.8908	4.3777	0.8524	0.7904	0.9048	0.7144	0.8155
PRACS	1.1803	3.4688	1.2773	3.8578	2.4461	1.9742	4.2221	2.4571	6.2161	3.7174	0.8166	0.7144	0.8804	0.6778	0.7723
MTF-GLP-FS	0.9239	2.3660	0.9808	2.7356	1.7516	1.8123	3.4496	2.2297	5.7208	**3.3031**	0.8792	0.8218	0.9238	0.8026	0.8568
MTF-GLP-HPM	0.9292	4.0148	1.2231	3.0609	2.3070	1.8555	5.1824	2.6120	6.4104	4.0151	0.8833	0.6698	0.9000	0.7890	0.8105
MTF-GLP-HPM-R	**0.9114**	2.4661	**0.9743**	2.6967	1.7621	**1.8121**	3.3953	2.2124	6.0986	3.3796	0.8800	0.8134	0.9239	0.8023	0.8549
AWLP	1.1846	4.6926	1.3424	3.8592	2.7697	2.3323	7.9280	2.9252	7.7668	5.2381	0.8516	0.6290	0.8832	0.7496	0.7783
MF	1.1392	4.0155	1.5506	3.3944	2.5249	1.9213	4.5587	2.7173	6.2918	3.8723	0.8638	0.7263	0.8804	0.7807	0.8128
HySURE	1.4664	4.3476	2.0749	5.1119	3.2502	2.9264	6.6950	4.4153	9.0468	5.7709	0.7943	0.5761	0.7475	0.4951	0.6532
SR-D	1.8476	4.2618	1.8825	4.8011	3.1982	2.4242	6.1149	3.0551	7.9927	4.8967	0.5734	0.5846	0.7313	0.5440	0.6083
TV	1.2624	2.8864	1.3163	3.1771	2.1605	2.2212	4.0400	2.5705	5.6672	3.6247	0.8291	0.7833	0.8911	0.7692	0.8182
HyperPNN	1.2245	2.2237	1.1864	3.3557	1.9976	2.8925	4.3058	2.6461	7.0296	4.2185	0.8777	0.8539	0.9247	0.8229	0.8698
HSpeNet	1.0354	2.0527	1.0674	2.7964	1.7379	2.0644	3.2157	2.2838	5.8462	3.3525	0.8674	0.8588	0.9136	0.8245	0.8660
DHP-DARN	1.9137	4.4280	1.7389	4.7617	3.2106	3.4673	5.1740	3.3833	9.1288	5.2883	0.8489	0.8234	0.9190	0.7934	0.8462
DIP-HyperKite	1.3510	2.0206	1.0995	2.8092	1.8201	3.6543	5.8268	2.7089	9.1932	5.3458	0.8419	0.8497	0.9205	0.8201	0.8581
Hyper-DSNet	0.9725	**1.8198**	0.9820	**2.4237**	**1.5495**	1.9162	**2.9638**	**2.1906**	7.4854	3.6390	**0.8833**	**0.8767**	**0.9303**	**0.8426**	**0.8832**
R-PNN	0.9884	2.2256	1.0581	2.6704	1.7356	1.9720	3.5728	2.3860	5.6797	3.4026	0.8712	0.8225	0.9172	0.8112	0.8555
PCA-Z-PNN	1.0824	2.9062	1.5575	2.7813	2.0819	2.1124	4.6125	3.1154	5.5297	3.8425	0.8507	0.7798	0.8647	0.8116	0.8267
ρ-PNN	1.0109	2.3268	1.0205	2.7308	1.7722	1.9625	3.5609	2.2439	**5.3396**	3.2767	0.8782	0.8279	0.9246	0.8177	0.8621

toolbox [40], we also provide the correlation-based index D_ρ proposed in [82]. Even so, no numerical index or combination of indices provides a fully reliable measure of spatial quality, and we will resort to visual inspection of sample results to get further indications on spatial quality.

C. Reduced-Resolution Quality Assessment

The numerical results obtained for the four RR datasets are summarized in Table VII. For each index, for each dataset, and on average (Avg.), the top 5 and the worst 5 methods are highlighted in green and red, respectively. The best one is in bold green. Before analyzing the results, focusing on DL methods, it is important to recall that supervised models are optimized for the RR case, while unsupervised ones (R-PNN, PCA-Z-PNN, and the proposed) are optimized for FR real data. Therefore, it is not surprising that several supervised techniques (e.g., HyperPNN, HSpeNet, and especially Hyper-DSNet) are on top. However, despite its unsupervised approach, the proposed solution also performs quite well. More specifically, it presents a good spectral behavior (SAM is the most appropriate index in this respect) and also a very good capacity to preserve spatial structures ($Q2^n$ is an index oriented to assess spatial correlation and contrast). It looks less competitive in terms of ERGAS, a band-wise normalized variant of the root-mean-square error, but still close to the top 5.

D. Full-Resolution Quality Assessment

FR numerical results are gathered in Table VIII using the same highlighting scheme of Table VII. The first five columns give the spectral quality index, D_λ, individually on the four test images and averaged on all of them. The next two groups of columns, instead, measure spatial quality by means of two alternative indices, D_S and D_ρ. In terms of spectral quality, the proposed method provides top scores, on par with TV, and much better than all other methods, including its ancestor R-PNN. This was our aim from the beginning, together with that of uniform quality across the whole spectrum, not caught by this index.

D_λ is a solid indicator of quality, as it is computed against a well-defined and unambiguous reference. Measuring spatial quality is much more challenging, which is why we consider two alternative indices. In fact, neither of them is without flaws. For example, D_S measures the distance between the PAN and the best (closest to the PAN) linear combination of the pansharpened spectral bands [89]. Therefore, if one band (among hundreds) is an exact copy of the PAN, then $D_S = 0$, no matter what the other bands look like. This happens for several entries in Table VIII which, far from signaling perfect pansharpening, correspond typically to degenerate solutions. While such cases are rarely seen with MS images, they occur more easily with HS images. Similar warnings apply to the second indicator. In fact, D_ρ is defined as the average of the local correlation coefficient between PAN and the pansharpened HS bands [82]. From a physical perspective, this metric evaluates the extent to which a $\sigma \times \sigma$ patch from any band of $\widehat{\mathbf{H}}$ can be linearly predicted from the corresponding patch in the PAN image. Consequently, D_ρ directly reflects the degree of spatial alignment between the pansharpened product $\widehat{\mathbf{H}}$ and the PAN image P at fine spatial scales. Therefore, it vanishes only when there is perfect correlation between all bands and

TABLE VIII
RESULTS AT FULL RESOLUTION. BEST SCORE IS IN BOLD GREEN, TOP-5 IN GREEN, AND WORST-5 IN RED

Method	D_λ					D_S					D_ρ				
	Cagliari	Udine	Ford	Tabasco	Avg.	Cagliari	Udine	Ford	Tabasco	Avg.	Cagliari	Udine	Ford	Tabasco	Avg.
(Ideal)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
EXP	0.0126	0.0077	0.0054	0.0106	0.0091	0.1306	0.1273	0.0704	0.1947	0.1308	0.7973	0.8839	0.7963	0.7972	0.8187
GSA	0.0227	0.0134	0.0078	0.0122	0.0140	0.0056	0.0074	0.0077	0.0184	0.0098	0.0593	0.6009	0.2075	0.0791	0.2367
BT-H	0.0633	0.0936	0.0195	0.0288	0.0513	**0.0000**	**0.0000**	**0.0000**	**0.0002**	**0.0001**	0.0579	0.0954	0.0544	0.0676	0.0688
BDSD-PC	0.0439	0.0256	0.0220	0.0487	0.0351	0.0024	0.0000	0.0001	0.0003	0.0007	0.1280	0.3878	0.1726	0.1192	0.2019
PRACS	0.0118	0.0155	0.0047	0.0088	0.0102	0.0083	0.0163	0.0114	0.0245	0.0151	0.1775	0.3252	0.3409	0.1903	0.2585
MTF-GLP-FS	0.0101	0.0054	0.0033	0.0065	0.0063	0.0214	0.0277	0.0174	0.0322	0.0247	0.0618	0.6000	0.2123	0.0831	0.2393
MTF-GLP-HPM	0.0147	0.0087	0.0067	0.0093	0.0098	0.0215	0.0284	0.0179	0.0389	0.0267	0.0479	0.1022	0.0438	0.0538	0.0619
MTF-GLP-HPM-R	0.0100	0.0057	0.0033	0.0074	0.0066	0.0217	0.0285	0.0176	0.0374	0.0263	0.0669	0.6050	0.2164	0.0950	0.2458
AWLP	0.0132	0.0097	0.0055	0.0092	0.0094	0.0251	0.0318	0.0206	0.0407	0.0295	0.0695	0.1778	0.0754	0.0933	0.1040
MF	0.0655	0.0392	0.0390	0.0331	0.0442	0.0389	0.0269	0.0250	0.0664	0.0393	**0.0476**	**0.0904**	**0.0391**	0.0585	**0.0589**
HySURE	0.1126	0.0618	0.0510	0.1158	0.0853	0.0022	0.0011	0.0016	0.0022	0.0018	0.1414	0.2276	0.0990	0.2688	0.1842
SR-D	0.0128	0.0087	0.0055	0.0115	0.0096	0.1316	0.1292	0.0720	0.1981	0.1327	0.8234	0.9061	0.8531	0.8560	0.8596
TV	**0.0042**	0.0028	0.0021	**0.0036**	**0.0032**	0.0586	0.0524	0.0358	0.0813	0.0570	0.2590	0.3294	0.2791	0.2001	0.2669
HyperPNN	0.0415	0.0139	0.0256	0.0248	0.0264	0.0036	0.0063	0.0040	0.0067	0.0051	0.1310	0.5080	0.2419	0.2625	0.2859
HSpeNet	0.0174	0.0084	0.0240	0.0154	0.0163	0.0101	0.0160	0.0141	0.0236	0.0159	0.1439	0.5338	0.2348	0.2558	0.2921
DHP-DARN	0.0832	0.0615	0.0467	0.0513	0.0607	0.0030	0.0053	0.0031	0.0130	0.0061	0.1494	0.4713	0.2106	0.2315	0.2657
DIP-HyperKite	0.0208	0.0125	0.0148	0.0146	0.0157	0.0042	0.0050	0.0043	0.0059	0.0048	0.2952	0.5270	0.2217	0.2765	0.3301
Hyper-DSNet	0.0119	0.0050	0.0082	0.0068	0.0080	0.0200	0.0246	0.0195	0.0346	0.0247	0.1957	0.5336	0.2276	0.3353	0.3230
R-PNN	0.0131	0.0091	0.0063	0.0074	0.0090	0.0168	0.0144	0.0129	0.0341	0.0195	0.0511	0.1366	0.0532	**0.0504**	0.0728
PCA-Z-PNN	0.0129	0.0041	0.0046	0.0080	0.0074	0.0152	0.0116	0.0088	0.0257	0.0153	0.0959	0.6828	0.7935	0.0792	0.4128
ρ-PNN	0.0043	**0.0027**	**0.0018**	0.0043	0.0033	0.0215	0.0168	0.0173	0.0380	0.0234	0.0790	0.1302	0.0743	0.0532	0.0842

the PAN. However, this is not desirable for bands that have no spectral overlap with the PAN and may differ significantly from it.

Keeping this in mind, we comment on the numerical results with some caution and await confirmation from the visual analysis to draw firm conclusions. In particular, the proposed method performs quite well in terms of spatial quality indices, especially when considering D_ρ. Together with the excellent results of D_λ, this speaks of a reliable and well-balanced method. This does not apply to TV, however, whose performance drops dramatically, suggesting questionable spatial quality. On the contrary, MF is characterized by excellent spatial quality indices but a very bad D_λ. Both these methods, and many others, seem unable to achieve a good balance between spectral and spatial fidelity. Overall, the proposed method compares quite favorably with all DL-based references. Only the R-PNN provides a better D_ρ on average, but this is also due to the perfect alignment between the metric and the spatial loss of the R-PNN, a property that does not hold with the new spatial loss used in the proposal.

E. Visual Results

The numerical results of Table VII provide indications on RR performances that are clear and reliable, although only proxies of the actual performances in the FR space. These latter are quantified directly in Table VIII. However, while D_λ captures spectral quality unambiguously, D_S and D_ρ provide more controversial and often contrasting indications on spatial quality. Therefore, we rely on the visual inspection of some sample results to gain a more solid insight into this latter point.

In Figs. 10 and 11, we show results for two close-ups from the FR Cagliari and the FR Tabasco datasets. Following [40], we show two different spectral triplets of bands for false-RGB composition, from the visible and NIR-SWIR spectral ranges. Given the large number of reference methods, we limit attention to the top-5 D_S and top-5 D_ρ methods (based on averages from Table VIII) plus TV, top-1 D_λ, for a total of 10 methods including the proposed one.[3] Together with the images, we also report the spectral (D_λ) and spatial (D_S and D_ρ) quality scores computed over all bands of the full scene.

From the visual inspection of images, we can make the following observations.

1) Results with low D_ρ are generally sharper, resembling closely the spatial structures of the PAN. This is true for scenes and in both the visible and NIR-SWIR spectral ranges. In particular, this is the case of BT-H, MF, MTF-GLP-HPM, R-PNN, and the proposed.
2) Results with low D_S reproduce the PAN spatial structures faithfully in the visible spectrum, but seem to fail outside that range. This is the case of BDSD-PC, HySURE, and DIP-HyperKite. BT-H seems an exception, characterized also by a low D_ρ; however, it presents a large spectral distortion.
3) The top-D_λ method, TV, whose spectral quality level is nearly the same as the proposed one, does not provide effective sharpening, providing blurred images.

Based on the above observations, we single out three methods, MTF-GLP-HPM, R-PNN, and our proposal, that

[3]The full set of results is available on the web repository at https://github.com/giu-guarino/rho-PNN

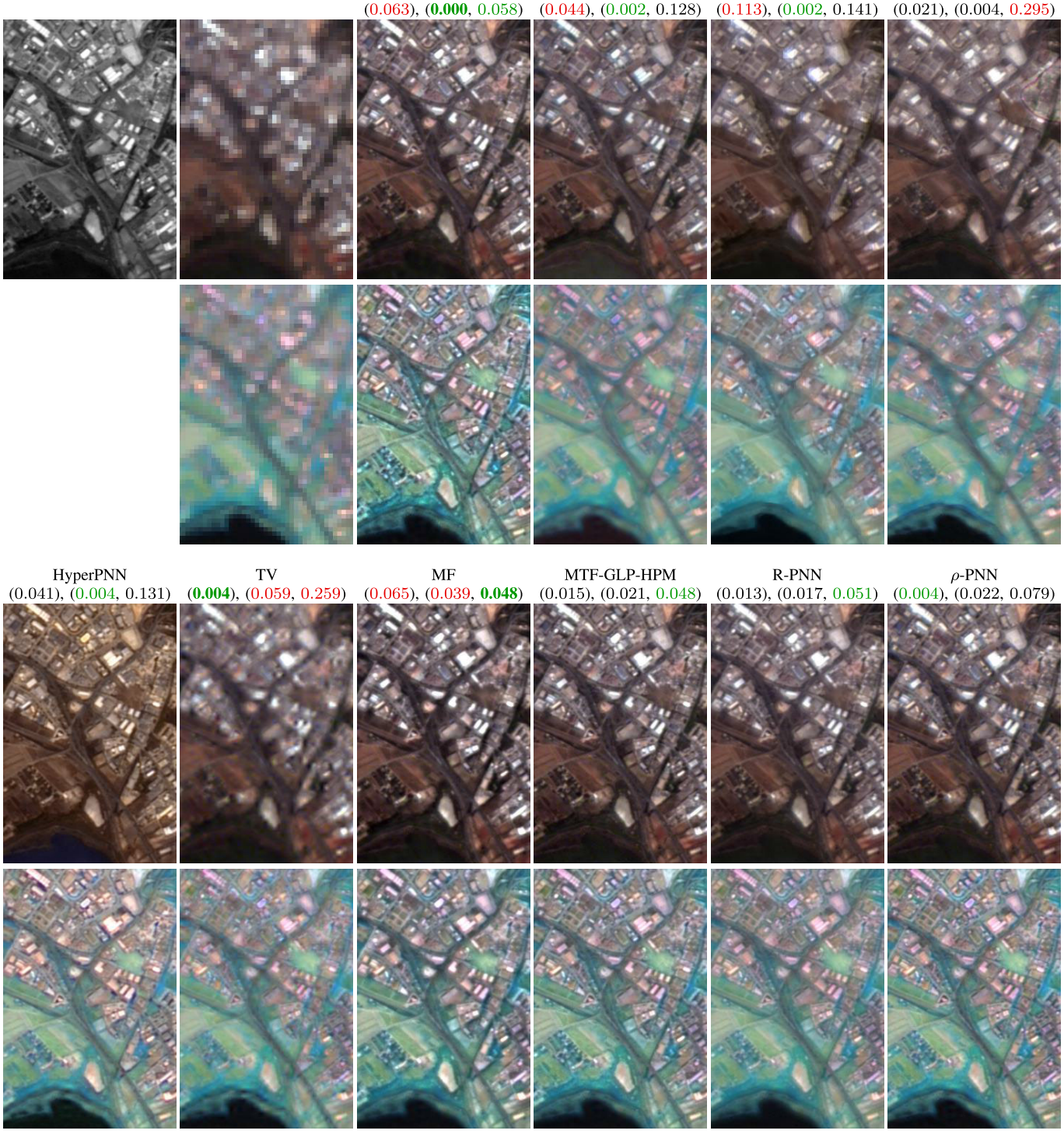


Fig. 10. Pansharpening results on the *FR Cagliari dataset* (240 × 360 close-up): the PAN image (top left) with paired lower resolution HS input (bands: 663, 560, and 466 nm) followed by a selection of pansharpening results. The input HS and all results are also displayed, on even rows, using a selection of bands from the NIR-SWIR range (1943, 1261, and 832 nm). For each result, the corresponding scores "(D_λ), (D_S, D_ρ)" are reported (from Table VIII).

appear to be both robust and effective, performing uniformly well on both scenes and all displayed spectral bands in terms of both spectral and spatial quality. Samples of these methods are shown close to each other, but the differences are so small that further examination seems like a futile exercise that could lead to claims related only to the choice of scene/bands/clips. On the other hand, all three methods have very similar spatial quality indicators. However, we point out that the proposed method has a much smaller average spectral distortion (0.0033, see Table VIII) than these competitors (0.0098 and 0.0090, respectively). Furthermore, only the proposed method ensures uniform quality across all bands, a feature not captured by the global indicators.

We complete this analysis by showing, in Fig. 12, some error maps obtained by subtracting the GT from the pansharpened image. In this case, we use the Udine dataset,

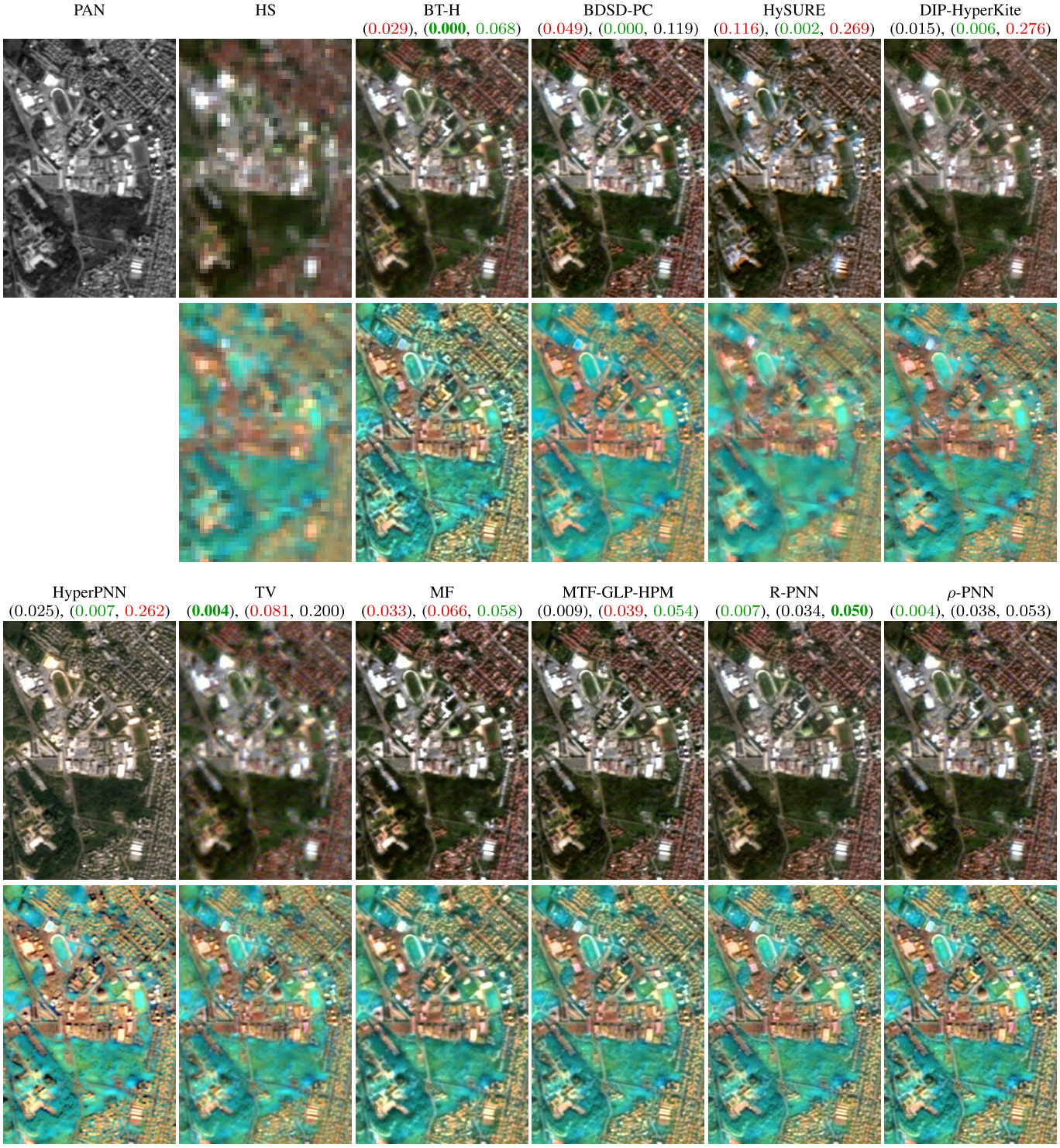
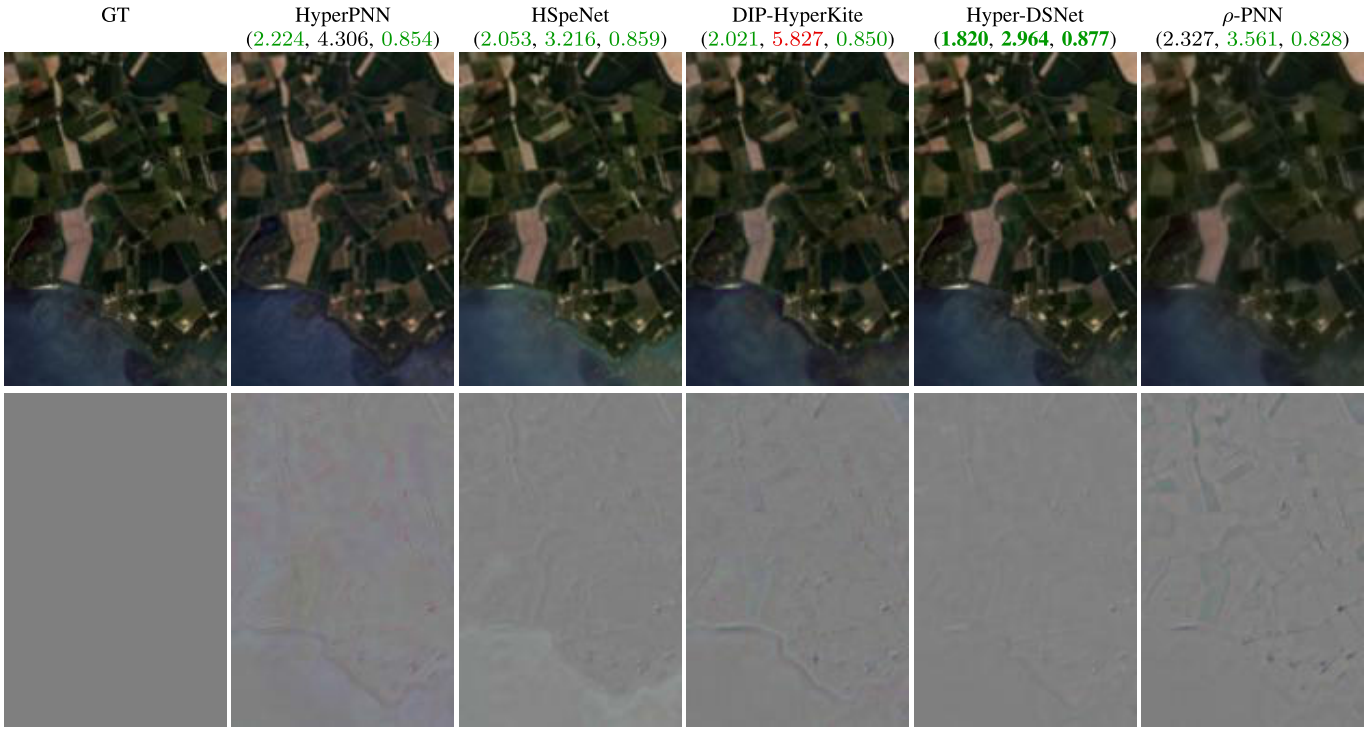
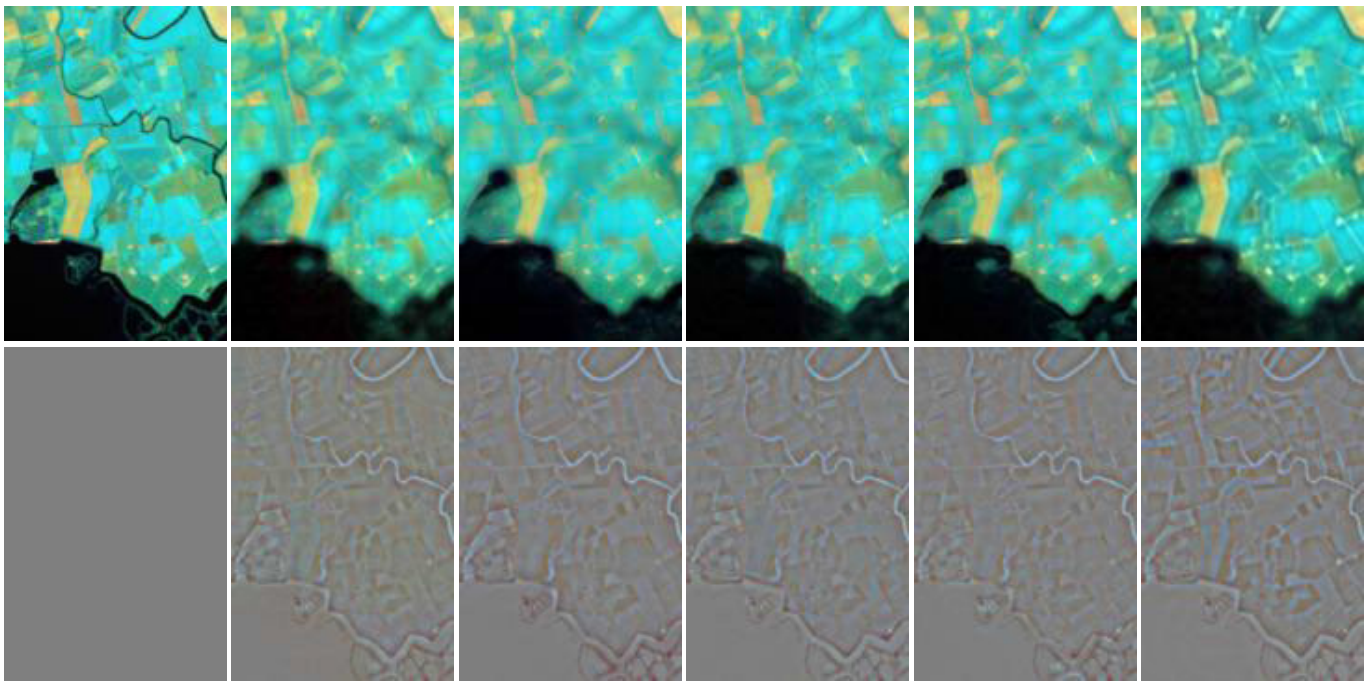


Fig. 11. Pansharpening results on the *FR Tabasco dataset* (240 × 360 close-up): the PAN image (top left) with paired lower resolution HS input (bands: 663, 560, and 466 nm) followed by a selection of pansharpening results. The input HS and all results are also displayed, on even rows, using a selection of bands from the NIR-SWIR range (1943, 1261, and 832 nm). For each result, the corresponding scores "(D_λ), (D_S, D_ρ)" are reported (from Table VIII).

where some interesting phenomena are visible, and work at reduced resolution, where the original HS component can be used as GT. To save space, only five methods are considered: the proposed one and the best four competitors according to the $Q2^n$ metric. For bands in the visible range (top), only small errors occur, and the error maps appear almost uniformly gray, except for HyperPNN, where some color aberrations are visible. As expected, larger errors appear for the NIR-SWIR bands (bottom) that do not overlap the PAN. Interestingly, some sharp structures visible in the GT (probably, waterways) are almost completely lost in the fused images, uniformly for all methods. On the other hand, such structures are hardly distinguishable in the visible bands and therefore in the PAN, suggesting



(a)



(b)

Fig. 12. Pansharpening results on the RR Udine dataset (120 × 180 close-up) for (a) visible bands and (b) NIR-SWIR bands. In both cases, GT and pansharpening results (top) with error maps (bottom). For each result, the corresponding scores "(ERGAS, SAM, and $Q2^n$)" are reported.

that the relevant information may be truly missing in this case.

F. Generalization Ability

As already said, ρ-PNN does not require any preliminary training phase but adapts on the fly to the target data. Therefore, we expect it to generalize smoothly to new sources, a key requirement of data-driven methods. To investigate this issue experimentally, we would need an alternative dataset, but we could find none available online, so we built a new one *ad hoc*, taking advantage of the good coverage and wide availability of Copernicus Sentinel-2 imagery. We found a Sentinel-2 image covering the same area as our PRISMA Kansas dataset with a temporal shift of one day. We used

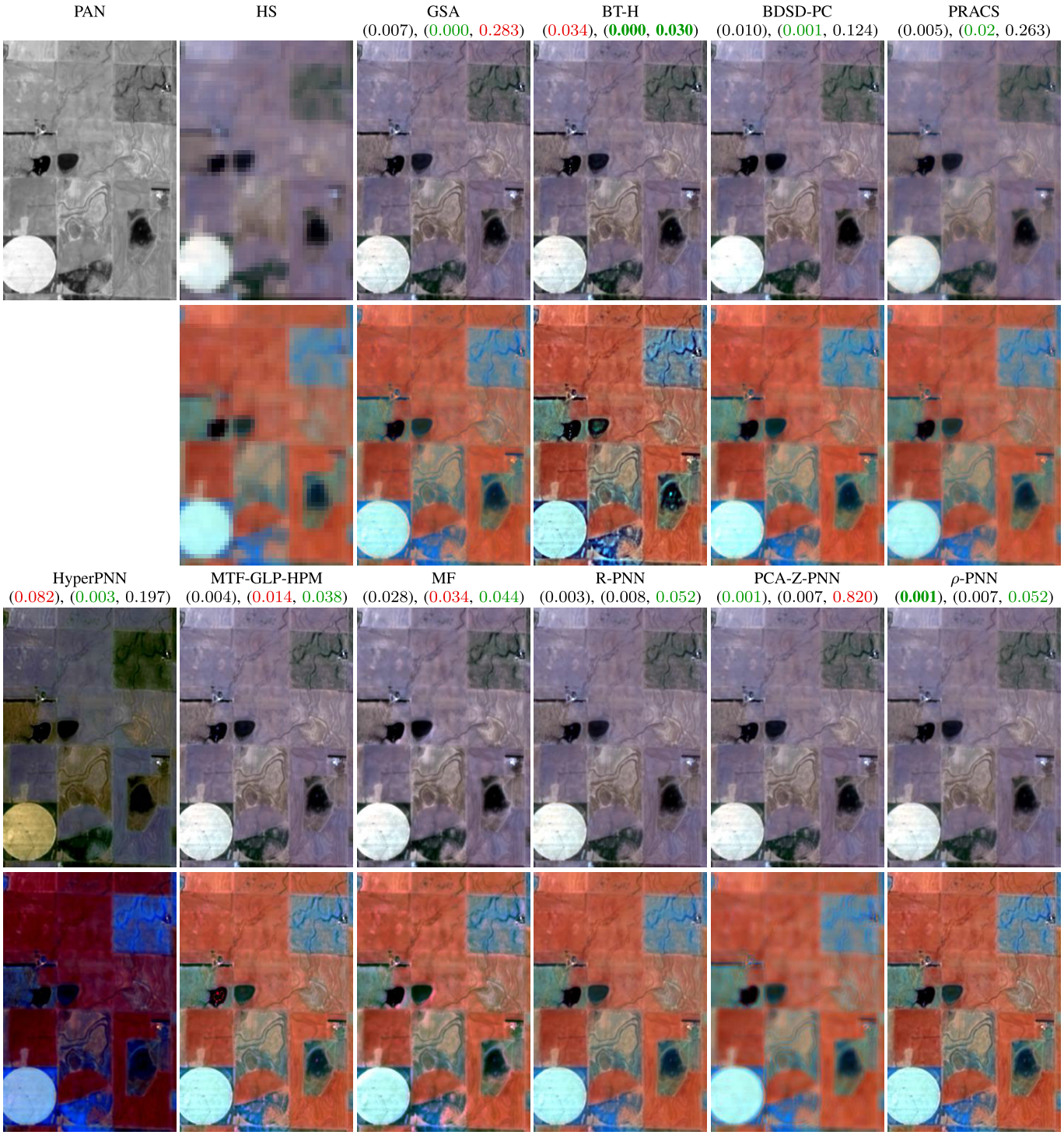


Fig. 13. Pansharpening results on the synthetic *Sentinel-2/PRISMA dataset* (240 × 360 close-up): the simulated PAN image obtained from Sentinel-2 RGB bands (top left) with paired reduced resolution PRISMA HS input (bands: 663, 560, and 466 nm) followed by a selection of pansharpening results. The input HS and all results are also displayed, on even rows, using a selection of bands from the NIR-SWIR range (1943, 1261, and 832 nm). For each result, the corresponding scores "(D_λ), (D_S, D_ρ)" are reported (from Table IX).

the RGB (665, 560, and 490 nm) high-resolution bands of the Sentinel-2 image to synthesize a 10-m resolution PAN, and co-registered it with the PRISMA HS dataset. Finally, to preserve the resolution ratio of 6 used in the toolbox [42] for all methods, the HS data were downscaled by a factor of 2, resulting in a spatial resolution of 60 m.

On this heterogeneous dataset, we repeated the same experiments carried out on the homogeneous PRISMA datasets; however, only at FR, for lack of sufficient data. The results are reported in Table IX and Fig. 13 with the by now usual format, where comparison methods are selected based on the results of Table IX. Both numerical and visual results confirm our expectations: ρ-PNN keeps working very well and is the only method that achieves a top-5 ranking for both spectral and spatial quality. On the contrary, several DL-based methods appear to struggle with these new data, especially in terms

TABLE IX
RESULTS AT FR ON THE *Sentinel-2/PRISMA Dataset*. BEST SCORE IS IN BOLD GREEN, TOP-5 IN GREEN, AND WORST-5 IN RED

Method	D_λ	D_S	D_ρ
(Ideal)	0	0	0
EXP	0.0039	0.0985	0.7919
GSA	0.0068	0.0002	0.2833
BT-H	0.0338	**0.0000**	**0.0298**
BDSD-PC	0.0095	0.0011	0.1236
PRACS	0.0047	0.0023	0.2629
MTF-GLP-FS	0.0019	0.0134	0.2820
MTF-GLP-HPM	0.0039	0.0140	0.0384
MTF-GLP-HPM-R	0.0019	0.0139	0.2861
AWLP	0.0034	0.0154	0.0626
MF	0.0278	0.0340	0.0439
HySURE	0.0553	0.0068	0.1323
SR-D	0.0043	0.1015	0.8624
TV	0.0018	0.0536	0.2117
HyperPNN	0.0824	0.0027	0.1967
HSpeNet	0.0920	0.0033	0.0799
DHP-DARN	0.1136	0.0035	0.3505
DIP-HyperKite	0.0044	0.0032	0.1657
Hyper-DSNet	0.0284	0.0121	0.1894
R-PNN	0.0032	0.0079	0.0516
PCA-Z-PNN	0.0013	0.0067	0.8200
ρ-PNN	**0.0013**	0.0071	0.0521

of spectral quality. This is especially obvious for the HyperPNN, with pansharpened images that are severely corrupted (see Fig. 13). Model-based methods, on the other hand, are not significantly affected by the data mismatch. In extreme synthesis, these results show that the proposed method keeps ensuring a very good and stable performance also on this new dataset, confirming the expected good generalization ability.

G. Discussion

This large body of experimental results fully confirms our expectations about the proposed method. For the sake of clarity, we summarize here its pros and cons, also with reference to competing methods.

From a structural point of view, ρ-PNN presents several qualifying points, some of which are shared only with its ancestor R-PNN and with PCA-Z-PNN.

1) It runs from scratch, without the need for prior training: this is obtained by means of a longer tuning run on the first band that affects only marginally the overall computational cost due to the very large number of bands.
2) It works with an arbitrary number of bands: as the algorithm works band-wise with no joint processing of the HS stack, the number of spectral bands does not need to be fixed to build the model.
3) It operates at FR with no resolution downgrading or simulation operations: as the proposed loss is fully unsupervised and based on input–output self-consistency, traditional resolution downgrade protocols (see [81]) to give rise to labeled, but synthetic, datasets are avoided.

Furthermore, in terms of performance

1) it is on par with the best SotA methods, both model-based and data-driven,
2) it guarantees a uniformly high spectral quality at all wavelengths, as only the TV method does, and
3) it shows excellent generalization ability.

On the downside, ρ-PNN has a relatively high computational cost compared to model-based techniques and even several data-driven techniques. This cost can be significantly reduced, however, by slightly relaxing the spectral quality constraints.

V. CONCLUSION

In this work, we have proposed a new pansharpening method aimed at ensuring a uniformly high fidelity over the whole spectral range. This is an important feature for HS images, where most bands are acquired at spectral frequencies not covered by the PAN, and may differ significantly from it. To this end, we leverage the rolling PNN architecture where pansharpening is carried out band-wise using cross-band transfer learning and a suitable zero-shot unsupervised tuning. From this base, we improve with two key innovations: a new loss, better adapted to the characteristics of HS images, that takes into account the correlation inversion phenomena that often occur in bands outside the visible range, and a hysteresis-based tuning scheme where the balance of spatial and spectral loss components changes dynamically to overcome unwanted low-quality solutions.

Extensive experimental assessment over a SotA benchmarking toolbox proves the proposed method to ensure excellent results both in terms of visual quality and according to the most credited quality indicators.

However, this latter point certainly deserves further research. To date, no spatial quality metric can be considered a fully reliable indicator of image quality at full resolution. Lacking a solid unambiguous reference, numerical assessment of performance remains inconclusive. This is the most urgent research topic in the pansharpening context, in our opinion, as it is intimately linked to the definition of an effective loss for unsupervised learning.

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